Realistic Mixing Miniature Scene Hyperspectral Unmixing: From Benchmark Datasets to Autonomous Unmixing

Chunyang Cui, Yanfei Zhong, *Senior Member, IEEE*, Xinyu Wang, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Mixed pixels that contain more than one material type are common in mid/low spatial resolution remote sensing imagery. Hyperspectral unmixing is aimed at decomposing the mixed pixels into endmembers and abundances. However, there are few datasets that are suitable for quantitatively evaluating unmixing accuracies, and the ground-truth abundances of the existing datasets are often generated in an approximate way. To address the lack of real unmixing datasets for quantitative evaluation, we built the realistic mixing miniature scenes (RMMS) dataset, which can be used to quantitatively evaluate the unmixing accuracy of different algorithms. The RMMS dataset consists of a simple mixture scene with homogeneous flat materials and a complex mixture scene with 3-D structural features. The features of the RMMS dataset also take point, line, and polygon characteristics into consideration, and the spectral similarity of the materials increases the challenge of the spectral unmixing. In the RMMS dataset, due to the multiscale observation characteristics of the spatiotemporal scanning modality, it can avoid the registration error between RGB and hyperspectral data, and it can ensure that the endmembers are pure pixels. Most of the autonomous hyperspectral unmixing algorithms focus on solving some of the unmixing problems and have difficulty achieving fully autonomous hyperspectral unmixing (FAHU). In this article, to overcome this shortcoming, a fully autonomous hyperspectral unmixing method called FAHU is proposed to take advantage of the spatial information. Some of the state-of-the-art autonomous hyperspectral unmixing algorithms are used to evaluate the performance with the RMMS dataset, and the experimental results show the advantages and disadvantages of the different autonomous unmixing algorithms.**

***Index Terms*— Abundance estimation (AE), endmember extraction (EE), fully autonomous, hyperspectral unmixing, real dataset, virtual dimensionality.**

I. Introduction

HYPERSPECTRAL images, which capture the spectral information in hundreds of narrow and contiguous bands, have advantages in many applications [1], [2]. However, due to the low spatial resolution of hyperspectral images, mixed pixels containing more than one material can seriously limit the accuracy of the quantitative interpretation of the images. Spectral unmixing, which is aimed at decomposing the mixed pixels into a set of pure materials, called endmembers, and their proportion in each mixed pixel, called abundances, is an effective way to solve the mixed pixel problem [3].

Spectral unmixing includes three basic problems: estimation of the number of endmembers (NOE), endmember extraction (EE), and abundance estimation (AE). The algorithms for autonomously solving the three basic problems can be categorized into six types: 1) NOE algorithms; 2) EE algorithms; 3) AE algorithms; 4) autonomous EE algorithms; 5) blind unmixing algorithms; and 6) fully autonomous hyperspectral unmixing (FAHU) algorithms. The NOE algorithms are aimed at determining the optimal NOE M in a hyperspectral image, which is crucial information for unmixing since the rank of the endmembers and the abundance matrices are heavily dependent on this. Eigen-analysis-based algorithms are widely used for NOE estimation [4], [5]. The EE algorithms focus on extracting a set of relatively pure spectra of the materials making up the hyperspectral image. Based on the relationship between the linear mixture model (LMM) and the convex sets, the endmembers are the most distinct spectra located at the vertex of the hyperspectral image data simplex. Most EE algorithms are based on the geometrical properties of the endmembers [6], [7], [8], [9]. The AE algorithms are aimed at estimating the proportions of the endmembers in each mixed pixel, which is usually considered as an inverse problem solved by a least-squares-based approach [10], [11]. Choosing one from each of the abovementioned categories of algorithms is one way to achieve hyperspectral unmixing. However, the three steps are independent and lack interaction, which causes error accumulation. The autonomous EE methods combine the NOE estimation and EE processes, thereby reducing the errors caused by the NOE estimation. A typical example is saliency-based autonomous endmember detection (SAED), which estimates the number and spectra of the endmembers jointly to reduce the cumulative error [12]. Blind unmixing is aimed at jointly estimating the endmember spectra and abundance maps, which avoids the cumulative error between the EE and AE processes [13], [14], [15]. Sparsity-promoting iterated constrained EE (SPICE) [16] is capable of estimating the NOE by adding the abundance sparsity based on

Manuscript received 30 September 2022; revised 9 November 2022; accepted 29 November 2022. Date of publication 13 January 2023; date of current version 8 February 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3903500, in part by the National Natural Science Foundation of China under Grant 42101327 and Grant 42071350, and in part by the LIESMARS Special Research Funding. *(Corresponding author: Yanfei Zhong.)*

Chunyang Cui, Yanfei Zhong, and Liangpei Zhang are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China (e-mail: cuichunyang@whu.edu.cn; zhongyanfei@whu.edu.cn; zlp62@whu.edu.cn).

Xinyu Wang is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China (e-mail: wangxinyu@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3236677

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

the iterated constrained endmember (ICE) [17] algorithm. Although SPICE can achieve fully autonomous unmixing, the results are not stable as the endmembers are randomly selected from the image, and it only uses the spectral information without considering the spatial–contextual information of the hyperspectral imagery.

Although there are many algorithms for hyperspectral unmixing, there are still many problems with the datasets used for hyperspectral unmixing.

1) For the synthetic datasets, although the ground truth is known, the hyperspectral images are mixed with a fixed mixing model, and the abundance maps are produced based on statistical assumptions. The advantage of the synthetic datasets is that the unmixing results can be accurately evaluated, but the mixed hyperspectral imagery is relatively simple when compared with real hyperspectral imagery.
2) With regard to airborne datasets, they are typically complex enough, but the ground truth is difficult to obtain and is based on manual interpretation, so that accuracy evaluation is hard to achieve.
3) With ground-based datasets, the ground truth is available, but the endmembers in the scenes are essentially the same substance, just in different colors, which means that the simulation is simple.

In this study, to achieve quantitative evaluation of hyperspectral unmixing and simulate both the simple and complex real mixing processes in an Earth surface system, the realistic mixing miniature scenes (RMMS) dataset based on a push-broom hyperspectral spectrometer (with spatiotemporal scanning modality) was built. The flowchart of this study is shown in Fig. 1. Specifically, the main contributions of this study can be summarized as follows.

1) The RMMS dataset simulates both simple and complex scenes. The RMMS dataset consists of a simple mixture scene with mixed homogeneous flat materials, i.e., moss, leaves, pebbles, and sticks, and a complex mixture scene consisting of succulents (true vegetation) and false succulents (false vegetation) with a 3-D structure. Compared with the existing ground-based datasets, the RMMS dataset has a more complex feature composition, with point, line, and polygon characteristics. The spectral similarity of the materials also increases the challenge of the hyperspectral unmixing.
2) The dataset contains the ground-truth NOE, the endmember spectra, and the abundances, which can be used to quantitatively evaluate the unmixing results. For the RMMS dataset, due to the multiscale observation characteristics of the spatiotemporal spectrometer, the ground-truth endmembers and abundances could be generated from the 16 × 0.125 mm/pixel hyperspectral image acquired by the same sensor at an adjacent time. This approach avoids the possibility of registration error between the RGB and hyperspectral images and, at the same time, avoids the need to select endmember spectra in an image of the same resolution, which ensures that the endmembers are pure pixels.

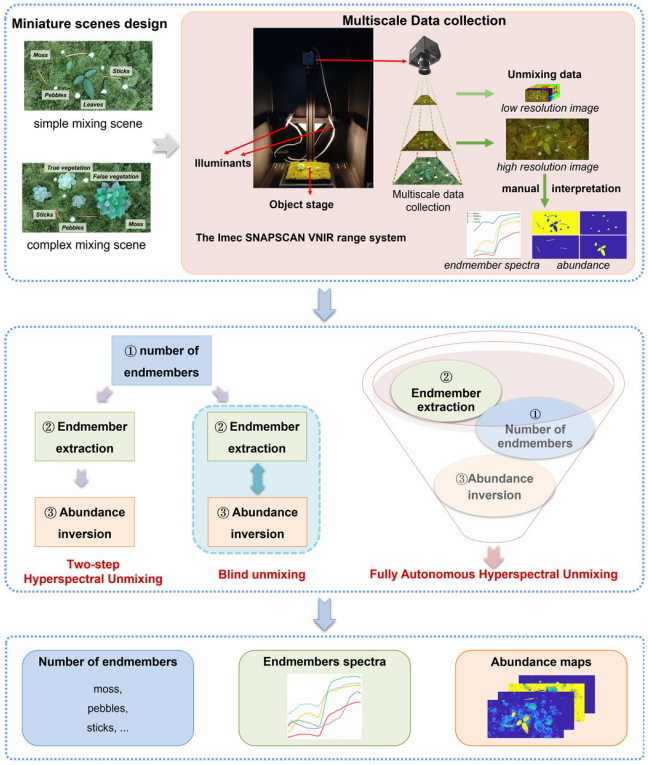


Fig. 1. Flowchart of this study.

3) The autonomous unmixing methods were tested using the benchmark RMMS dataset. There are many state-of-the-art algorithms for hyperspectral unmixing, which can solve some or all of the problems of hyperspectral unmixing. Based on this, benchmark tests were conducted on the RMMS dataset. In order to achieve fully autonomous unmixing and take advantage of the spatial information, the FAHU method based on SAED and spatial group sparsity regularized nonnegative matrix factorization (SGSNMF) is presented. The SAED method can estimate the NOE and the endmember spectra at the same time, so that it can provide the initialization for SGSNMF, which is used for the AE. The FAHU method can solve the three problems of hyperspectral unmixing, without the need for any prior knowledge, and can take advantage of the spatial information of the hyperspectral imagery.

The rest of this article is organized as follows. Section II reviews the datasets and algorithms for hyperspectral unmixing. In Section III, the data collection process of this study is described. The results of the experiments and a discussion are provided in Section IV. Finally, our conclusions are presented in Section V.

II. Background

This section introduces the published datasets and algorithms for hyperspectral unmixing in detail.

A. Public Datasets for Hyperspectral Unmixing

As illustrated in Table I, the synthetic datasets for quantitatively evaluating unmixing accuracy are very few in number,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF THE DATASETS COMMONLY USED FOR EVALUATING UNMIXING ACCURACY

Type	Dataset	Scene type	Mixing model			Ground-truth reference		
			LMM	NLM	Unknown	VD	Endmembers	Abundances
Synthetic	Simulated data cube [17]	mineral	✓			✓	✓	A
	Fractal 1 data [18]	mineral	✓			✓	✓	A
Airborne	Samson [19]	rural			✓	×	✓	C
	Jasper Ridge [19]	rural			✓	×	✓	C
	Urban [19]	urban			✓	×	✓	C
	Cuprite [20]	mineral			✓	×	✓	×
	Washington DC Mall [21]	urban			✓	×	✓	×
	Pavia University [22]	urban			✓	×	✓	×
Ground-based	Checkerboards [23]	colour	✓			✓	✓	B
	Mixed quartz sands [23]	sands		✓		✓	✓	B

A: computer simulation; B: estimation by a third-party unmixing method; C: manual interpretation via a high-resolution RGB image;
LMM: linear mixture model; NLM: nonlinear mixture model; VD: virtual dimensionality.

which is mainly because the abundances are fractional numbers, and thus, the ground truth of the abundances cannot be quantitatively interpreted by visual means. For the existing datasets, the ground-truth abundances have generally been estimated in an approximate way, i.e., computer simulation, estimation by a third-party unmixing method or manual interpretation via high-resolution RGB imagery. According to the method of data acquisition, as shown in Table I, the datasets can be roughly divided into three main classes with the following characteristics.

1) Synthetic datasets have been widely used to quantitatively evaluate unmixing accuracy, e.g., simulated data cubes [18] or the Fractal 1 dataset [19]. The endmember number, endmember spectra, and abundances are generated through computer simulation, so that the ground truth is known. However, synthetic datasets are usually generated based on a specific mixing model, e.g., the LMM, and the abundance maps have a fixed statistical distribution, e.g., a Dirichlet distribution. As such, these datasets are not suitable to verify the generalization ability of unmixing algorithms in real and complex scenes.
2) Airborne datasets are collected by airborne hyperspectral imagers. Well-known examples of airborne hyperspectral datasets are the Samson [20], Jasper Ridge [20], Urban [20], Cuprite [21], Washington dc Mall [22], and Pavia University [23] datasets. Generally speaking, these airborne datasets are commonly used for classification tasks, but only pixel-level labels are available for the visual interpretation, and the ground-truth abundances are unknown, as well as the exact NOE, due to the large land-cover area. As a compromise, the currently used ground-truth abundance maps for the Samson, Jasper Ridge, and Urban datasets were approximated based on a third-party unmixing method and are thus not real. Therefore, the ground-truth abundance maps cannot be used to accurately evaluate the unmixing performance.
3) Ground-based datasets are built by the use of a ground-based line-scanning hyperspectral imager and a high-resolution RGB sensor, e.g., the checkerboard and mixed quartz sand scenes in the dataset released by Zhao et al. [24]. Since the scene coverage is small, the ground-truth NOE and the endmembers can be interpreted via field survey, and the abundances can be generated by downsampling a manually labeled high spatial resolution classification map based on the corresponding high-resolution RGB image. However, the scenes in the ground-based dataset released by Zhao et al. [24], i.e., the checkerboard and mixed quartz sand scenes, are relatively simple, with a monotonous material type. As such, these scenes cannot fully simulate the real scenes of Earth surface systems.

The synthetic datasets used for hyperspectral unmixing can be used to quantitatively evaluate the unmixing performance, but the simulated hyperspectral images are relatively simple when compared with real scenes. Airborne data are ideal data for hyperspectral unmixing, but it is difficult to obtain a reliable ground-truth reference. Ground-based data can be used to design the mixing scene and accurately provide reliable ground-truth information, but the materials in the designed scene are relatively simple.

B. Algorithms for Hyperspectral Unmixing

There are many state-of-the-art autonomous unmixing algorithms for hyperspectral unmixing. There are three basic problems for hyperspectral unmixing, which are the estimation of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SUMMARY OF THE TYPICAL ALGORITHMS FOR AUTONOMOUS HYPERSPECTRAL UNMIXING

Type	Features	Scientific problem			Typical algorithms
		NOE	EE	AE	
①	Unsupervised estimation of the NOE. Lacks interaction with the other two problems.	✓			HySime [3], NWHFC [4], GENE [24]
②	Unsupervised EE. The NOE affects the results.		✓		VCA [5], OSP [8], N-FINDR [6], MVSA [7], entropy-based convex set optimization [25]
③	Unsupervised AE. Isolated AE.			✓	NCLS [8], FCLS [9], EVLMM[29], GBM[30], PPNMM[31]
①+②	Unsupervised extraction of the NOE and endmember spectra. The results depend on the endmember spectra.	✓	✓		ATGP/MOCA [33], SAED [10]
②+③	Unsupervised extraction of the endmember spectra and abundances. The NOE affects the results.		✓	✓	MVCNMF [12], SGSNMF [13], uDAS [14], CSsRS [36], CyCU-Net[37], JMnet [38]
① +②+③	Fully autonomous unmixing. No prior knowledge of the hyperspectral imagery is required.	✓	✓	✓	FAHU [10], [13], SPICE [15],

① number of endmembers (NOE); ② endmember extraction (EE); ③ abundance estimation (AE)

the NOE, EE, and AE. The typical unsupervised algorithms for solving these three problems are summarized in Table II. The algorithms can be categorized into six types: algorithms which only estimate the NOE (marked as ①); algorithms which only extract the endmember spectra (marked as ②); algorithms which only inverse the abundances (marked as ③); algorithms which simultaneously estimate the NOEs and the endmember spectra (marked as ① + ②); algorithms which simultaneously estimate the endmember spectra and abundances (marked as ② + ③); and algorithms that can achieve FAHU (marked as ① + ② + ③).

1) Number of Endmember Estimation: The optimal NOE, i.e., M, is a crucial first step in hyperspectral unmixing. The eigen-analysis-based NOE estimation algorithms are commonly used for this task, e.g., the hyperspectral signal subspace identification by minimum error (HySime) method [4] and the noise-whitened Harsanyi–Farrand–Chang (NWHFC) method [5]. HySime is a fully autonomous method, where the NOE equals the dimension of the best signal subspace, which is determined by minimizing the projection error. The NWHFC method removes the noise effect through noise-whitening preprocessing and uses the Neyman–Pearson detector to find the difference between the sample correlation eigenvalues and sample covariance eigenvalues. This means that the virtual dimensionality, defined as the minimum number of distinct spectral signals to represent the hyperspectral image, is the NOE. The afore-mentioned algorithms can not only solve the hyperspectral unmixing problem but can also be used for target detection, classification, and so on [4]. Based on the fact that all the image pixels lie in the convex hull or the affine hull of the endmembers, the geometry-based estimation of the number of endmembers (GENE) algorithm [25] determines the NOE by a Neyman–Pearson detector. GENE-based algorithms include the GENE-convex hull (GENE-CH) algorithm and the GENE-affine hull (GENE-AH) algorithm. However, these methods only focus on estimating the NOE, without any interaction with the EE and AE, and if the NOE is not exact for the hyperspectral unmixing, the error in the NOE will have an effect on the final hyperspectral unmixing accuracy.

2) EE and AE: When the NOE is known as a prior, the traditional LMM-based hyperspectral unmixing problem can be divided into two subproblems—EE and AE—which are then solved separately.

The geometrical property of the hyperspectral data is the main consideration for EE, which can be categorized into two main types: hyperspectral images with pure pixels and hyperspectral images without pure pixels. For the pure pixel assumption, the EE is aimed at collecting a set of pure pixels from the image, by methods such as vertex component analysis (VCA) [6], N-FINDR [7], orthogonal subspace projection (OSP) [9], and entropy-based convex set optimization [26]. VCA [6] determines the endmembers by iteratively projecting the image pixels on to a direction orthogonal to the subspace spanned by the determined endmembers, and the largest projection of the corresponding pixel is selected as the new endmember. The iteration stops when all the endmembers are exhausted. For N-FINDR [7], it is aimed at finding the set of pixels with the largest simplex volume. The basic concept of OSP [9] is to project each pixel vector onto an orthogonal subspace until all the desired endmembers are found. Entropy-based convex set optimization [26] improves the EE by exploiting the entropy as the spatial information and convex set optimization as the spectral information. For the case of there being no pure pixels in the hyperspectral image, pseudo-endmembers can be generated by describing a simplex, by methods such as the minimum volume simplex analysis (MVSA) algorithm [8] and the minimum-volume enclosing simplex (MVES) algorithm [27]. The MVSA algorithm [8] is based on the nonpure pixel assumption and fits a minimum volume simplex to the hyperspectral data by constraining the abundance fractions to belong to the probability simplex. The MVES method [27] is based on Craig's criterion [28] and uses linear programs (LPs) to solve the optimization problem. The EE plays an important role in hyperspectral unmixing. Thus, the accuracy of the EE directly influences the accuracy of the AE.

The LMM is the most popular mixing model and is based on the assumption that the light only interacts with a single material before being received by the sensor. The AE can then be considered as an inverse problem and solved by least-squares based approaches, such as the nonnegative constrained least squares (NCLS) method [10] and fully constrained least squares (FCLS) [11]. Based on the nonlinear mixing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

models, there are also many unmixing algorithms, such as endmember variable LMM (EVLMM) [29], generalized bilinear model (GBM) [30], polynomial postnonlinear mixing model (PPNMM) [31], and so on.

3) Autonomous EE: The traditional EE algorithms require the NOE as prior knowledge, but this can cause cumulative error for the NOE. Autonomous EE algorithms can solve this problem by estimating the NOE and the endmember spectra at the same time. For example, Chang et al. [32] interpreted the connection between the maximum orthogonal complement algorithm (MOCA) [33] and the automatic target generation process (ATGP) method [34] and introduced Neyman–Pearson detection into the MOCA as the detector, so that the ATGP/MOCA method can estimate the NOE and extract the endmember spectra at the same time. It can be found that this method takes advantage of the geometric characteristics of the endmembers, but it ignores the spatial properties of the endmembers. In order to overcome this, SAED [35] exploits the connection between the NOE, the endmember spectra, and the abundances, which can take advantage of the spatial–contextual information via saliency analysis and can achieve autonomous EE.

4) *Blind Unmixing:* The EE results can have a great effect on the precision of the estimated abundances. Blind source separation (BSS) can be used to avoid the disadvantages mentioned above, where the observed signals, mixing matrix, and source signals can be considered as the mixed pixels, endmembers, and abundances in the hyperspectral unmixing, respectively [13], [14]. Therefore, an ideal approach is to obtain the endmember spectra and abundances at the same time. In view of the nonnegative properties of abundances and endmember spectra, nonnegative matrix factorization (NMF) is an ideal algorithm for linear unmixing. However, the basic NMF methods may not obtain a unique solution when only using the nonnegativity constraint. Hence, it is necessary to introduce additional auxiliary constraints, and many constrained NMF-based blind unmixing algorithms have been proposed. Minimum-volume CNMF (MVCNMF) [13] considers the simplex volume, and for ICE [17], the difference between endmembers is controlled in the iterative process. SGSNMF [14] incorporates a spatial group sparsity regularizer constraint into the NMF-based unmixing process. The correntropy-based spatial–spectral robust sparsity-regularized (CSsRS) unmixing model [36] uses a correntropy-based metric to reduce the impact of noise and improve the robustness. With the ongoing development of deep learning, autoencoders have also been used to deal with the blind hyperspectral unmixing problem. In [15], an untied denoising autoencoder with sparsity for spectral unmixing (uDAS) was proposed. Gao et al. [37] proposed the cycle-consistency unmixing network (CyCU-Net) using two cascaded autoencoders and self-perception loss to improve the inversion of the detailed information. Min et al. [38] considered the distribution similarity of the original and reconstructed images by introducing the Wasserstein distance and feature matching as regularization terms of the joint metric neural network (JMnet) for hyperspectral unmixing.

5) Fully Autonomous Hyperspectral Unmixing: Although autonomous EE and blind unmixing can solve two of the three main problems, the processes are still independent of each other and lack a feedback mechanism, which results in a large amount of accumulated errors in the hyperspectral unmixing. The SPICE algorithm [16] can solve the hyperspectral unmixing problem in a fully autonomous manner. SPICE is an extension of the ICE algorithm, which incorporates sparsity-promoting prior constraints in the abundance matrix to find the correct NOE. The initial endmembers are randomly selected from the hyperspectral image and are overestimated. Based on the sparsity promotion, the abundances of the unnecessary endmembers are driven to zero and pruned based on the predefined threshold. Although SPICE does not require prior knowledge of the NOE, the initial endmembers are randomly chosen from the hyperspectral image, which means that the endmembers with fewer pixels are less likely to be selected, and it does not take advantage of the spatial information. In this article, in order to overcome these issues, the FAHU method based on SAED and SGSNMF is presented. The SAED method can establish the connection between the NOE, the endmember spectra, and the abundances, which can provide a relatively stable initialization for SGSNMF with regard to the NOE and the endmember spectra. Furthermore, by considering the spatial–contextual information, SGSNMF can inverse relatively accurate abundances. The FAHU method can solve the three main problems of hyperspectral unmixing, with no need for any prior knowledge, and it can also take advantage of the spatial information of the hyperspectral imagery.

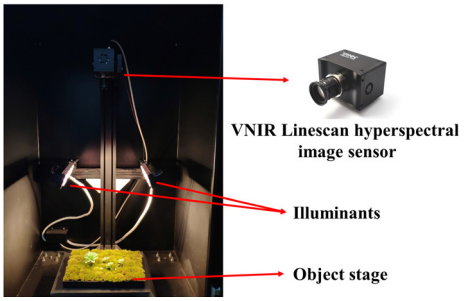


Fig. 2. Imec SNAPSCAN VNIR range system.

III. RMMS Dataset

The benchmark RMMS dataset was built using an Imec SNAPSCAN visible and near-infrared (VNIR) range system. The dataset contains two scenes. The first scene features simple mixing, and the other scene contains high-order interactions and true and false vegetation and is thus more complex. In this section, the design of the two scenes is described in detail.

A. Data Collection System

The data were collected using Imec's SNAPSCAN VNIR range system, which is a useful tool for hyperspectral imaging application research. As shown in Fig. 2, the data collection system contains three main parts: 1) the VNIR Linescan hyperspectral image sensor; 2) the halogen (cool fit) illuminants; and 3) the object stage. The sensor is designed to work in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

TABLE III
MAIN SPECIFICATIONS OF THE VNIR LINESCAN HYPERSPECTRAL IMAGE SENSOR IN THE SNAPSCAN SYSTEM

Specification	**SNAPSCAN system**
Spatial resolution	up to 3650 × 2048 px
Spectral resolution	150+ bands
Spectral range	470–900 nm
SNR	> 100–200, flat SNR over the spectral range

VNIR spectral range, i.e., 470–900 nm. The system can collect high-level data with a high spatial and spectral resolution. Within as little as a few hundreds of milliseconds, high-quality hypercube datasets can be created with a superb signal-to-noise ratio (SNR). The great advantage of this device is that it can obtain different resolutions of the same scene. The main specifications of the Imec VNIR Linescan hyperspectral image sensor are listed in Table III.

B. Data Collection and Processing of the SMScene

In order to simulate a simple mixing scene, called SMScene, the scene was designed to be homogeneous and flat, with no intimate mixing. There are four objects in the scene: moss, pebbles, sticks, and leaves. The locations and the number of each type of object are shown in Fig. 3(b). Point, line, and polygon features are the basic elements of polygon space, so stones, sticks, leaves, and moss are used to represent these features. There are four endmembers in this scene. Although both the moss and leaves are vegetation, they are two different endmembers, which increases the complexity of the designed scene.

By taking advantage of the fact that the hyperspectral imaging system can obtain different-resolution images of the same scene, two different-resolution images of the SMScene were collected. The resolution of the high-resolution image, as shown in Fig. 3(b), which was used to obtain the abundance ground truth, is 0.125 mm/pixel, with a 1024 × 2048 × 3 image size. As the spatial resolution of the imaging system is high, the scale difference between the two resolutions is also high. Therefore, the mixed hyperspectral image, as shown in Fig. 3(a), has a spatial resolution of 16 × 0.125 mm/pixel, with a 64 × 128 × 147 image size, so that the scale factor for the two different-resolution images is 16. The integration times of the different resolutions in the scene were set to be the same, so that the only variable was the resolution of the scene.

After collecting the data, some data preprocessing was required, which involved removing noisy bands and manually interpreting the ground truth. The total number of bands for the original hyperspectral images was 147. After the noisy bands were removed, there were 132 bands in the final images.

Manual interpretation of the ground truth of the endmember spectra and abundances was also conducted. In order to obtain the true abundances of the endmembers, the regions of interest were marked on the high-resolution image by manual interpretation. The classification images of the SMScene are shown in Fig. 3(c). By counting the proportion of each endmember, it can be found that the moss makes up 94% and is the most widely distributed material, and the sticks make up 0.7% and are the least common material in the scene. The pebbles and leaves make up 1.4% and 3.9%, respectively. The true abundances of the endmembers were obtained by computing the fraction of each endmember in a 16 × 16 pixel block in the high-resolution image. The results are shown in Fig. 3(d). Knowing the true abundances of the endmembers makes it possible to select the pure endmember spectra. The pure pixels of each endmember, for which the abundance equals one, were collected from the low-resolution image and averaged. The true endmember spectra for the SMScene are shown in Fig. 3(e). For the spectral variability of the features, finding the location of the extracted endmember spectra is a good way to analyze the results of the endmember spectra extraction. A metric combining the spectral angle distance (SAD) with the Euclidean distance was used in this study. This can be expressed as

$$W_{i,j} = \arccos\frac{\hat{\mathbf{a}}_i^T \mathbf{a}_j}{\|\hat{\mathbf{a}}_i\|_2 \|\mathbf{a}_j\|_2} + \left\| \frac{\hat{\mathbf{a}}_i - \mathbf{a}_j}{L} \right\|_2 \tag{1}$$

where $\mathbf{a}_i$ is the ground-truth spectral signature of the $_{i\text{th}}$ endmember, and $\hat{\mathbf{a}}_j$ is the spectral signature of the hyperspectral image. L is the number of spectral bands. The locations of the endmember spectra are marked with the red circles in Fig. 3(f).

C. Data Collection and Preprocessing of the CMScene

In the simulation of the complex scene, called CMScene, we placed five types of objects: moss, pebbles, sticks, succulents (true vegetation), and false succulents (false vegetation). In Fig. 4(b), it can be seen that the locations of the objects are clear, and that the NOE equals the number of features, i.e., five. In this scene, the basic elements of polygon space, i.e., point, line, and polygon features, are also taken into consideration. There are two aspects to the complexity of this scene. One is that, for the sticks endmember, there are no pure pixels in the scene, and the other is that the scene contains both true and false vegetation, which are difficult to distinguish visually.

For the CMScene image, these data were also collected by the use of the Imec SNAPSCAN VNIR range system. In order to obtain the ground-truth abundances, the high-resolution image was collected with a spatial resolution of 0.125 mm/pixel and a 1024 × 2048 image size, as shown in Fig. 4(b). To make the low-resolution image more mixed, the scale factor of the two images was 16. Therefore, the resolution of the mixed hyperspectral image is 16×0.125 mm/pixel, with 64 × 128 pixels, as shown in Fig. 4(a). The integration times of the different resolutions in this scene were set to be the same, so that the only variable was the resolution of the scene.

The first preprocessing operation was to remove the noisy bands, so that the final image for unmixing contained 132 bands (from 147 initial bands).

The second preprocessing operation was to obtain the ground-truth endmember abundance maps and spectra. As with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

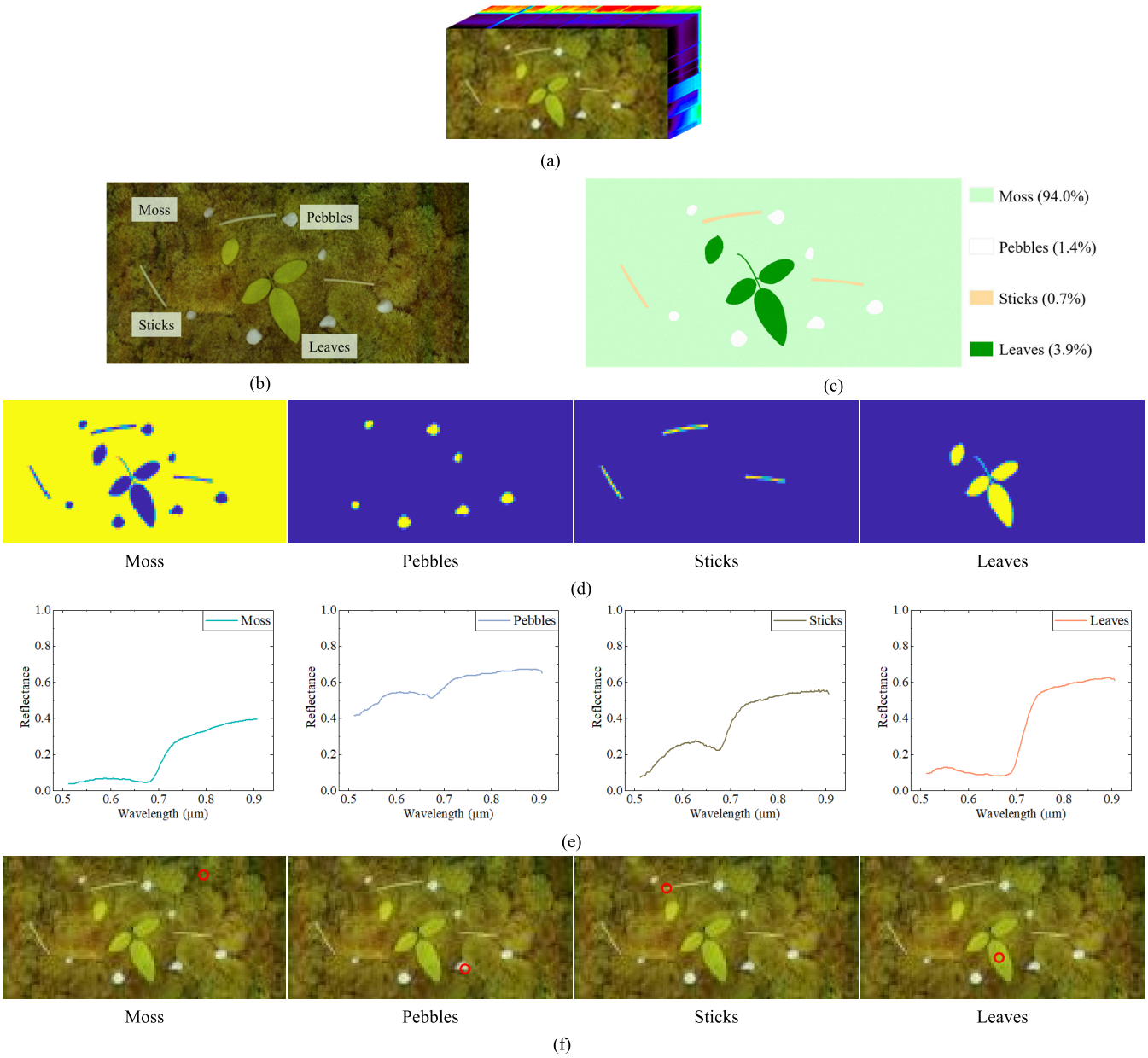


Fig. 3. Collected and processed ground-truth images for the SMScene image. (a) Mixed hyperspectral image data cube (64 × 128 × 132). (b) High-resolution image (1024 × 2048 × 3). (c) Manually interpreted classification map. (d) Ground-truth abundance maps. (e) Ground-truth endmember spectra. (f) Location of the endmember spectra.

the high-resolution image, the ground-truth distribution of the materials was obtained by manual interpretation, as shown in Fig. 4(c). We also analyzed the proportion of each endmember, and it was found that the moss makes up of 81.5% and is the most widely distributed material. The sticks make up of 0.5%, occupying the smallest proportion in the scene. The pebbles make up of 0.8%. The proportions of the true and false vegetation are similar, at 6.9% and 10.3%, respectively. Based on the classification image, the ground-truth abundance maps, as shown in Fig. 4(d), were obtained by computing the proportion of each endmember in a 16 × 16 pixel block. The ground-truth endmember spectra were obtained based on the abundance maps of the endmembers. The endmember spectra are equal to the average spectra of the pure pixels of the low-resolution image, for which the abundance equals one. As there was no pure spectrum for sticks in this scene, the pure spectrum of the sticks in the SMScene was considered as the pure spectrum in this scene. The wavelength of the endmember spectra references is shown in Fig. 4(e). In order to verify the correctness of the reference endmember spectra, the metric combining the SAD with the Euclidean distance, as expressed in (1), was also used in this CMScene. The locations of the endmember spectra are marked with the red circles in Fig. 4(f).

IV. Experiments and Analysis

In this section, we describe how the FAHU algorithm was compared with the state-of-the-art hyperspectral unmixing methods using the RMMS dataset. The main purpose was

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

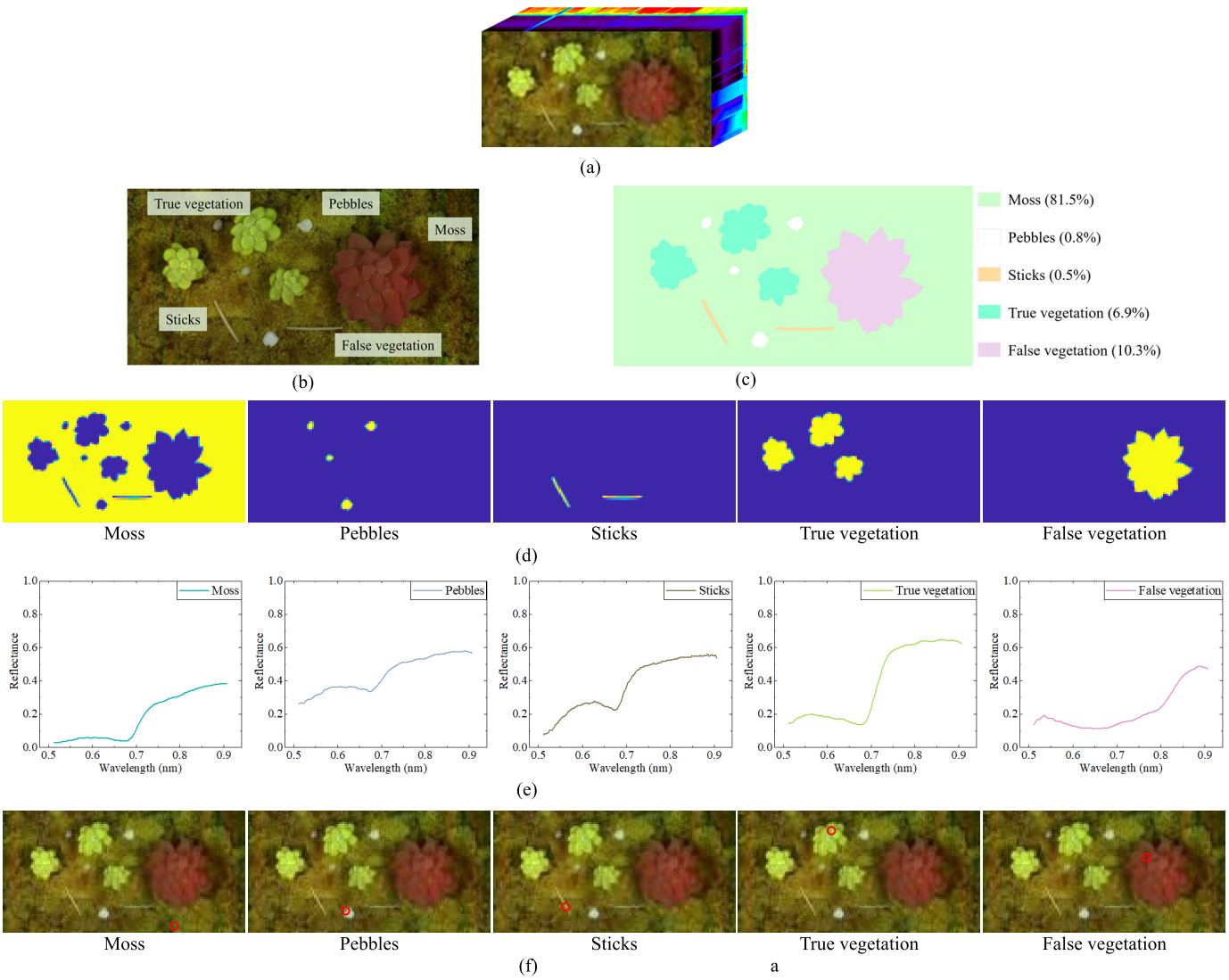


Fig. 4. Collected and processed ground-truth images for the CMScene image. (a) Mixed hyperspectral image data cube (64 × 128 × 132). (b) High-resolution image (1024 × 2048 × 3). (c) Manually interpreted classification map. (d) Ground-truth abundance maps. (e) Ground-truth endmember spectra. (f) Location of the endmember spectra.

to analyze the performance of the proposed algorithm in the NOE, EE, and AE tasks.

A. Data Description

1) SMScene: This full scene of the simulated SMScene [as shown in Fig. 3(a)] is of 64 × 128 pixels in size, with four endmembers—moss, pebbles, sticks, and leaves—containing point, line, and polygon features. The endmember abundance ground truth was manually interpreted based on the high-resolution image shown in Fig. 3(d). The endmember spectra were collected as the average of the pure pixels in the mixed image shown in Fig. 3(e).

2) CMScene: This is a full scene of the simulated CMScene, as shown in Fig. 4(a). The image is of 64 × 128 pixels in size, with five endmembers, which are moss, pebbles, sticks, true vegetation, and false vegetation. The ground-truth abundance maps [as shown in Fig. 4(d)] are illustrated based on the high-resolution image, which is 16 times larger than the mixed hyperspectral image. The ground-truth endmember spectra were determined according to the pure pixels of the mixed hyperspectral image, as shown in Fig. 4(e).

3) SMScene-sub: This image is a subset of the SMScene image [as shown in Fig. 5(a)], with only three endmembers of moss, pebbles, and leaves, and is without the linear feature of sticks. The image is 45 × 45 pixels in size, and the corresponding abundance ground truth was obtained based on the SMScene high-resolution image with 720 × 720 pixels. The spectra of the three endmembers are the same as the corresponding spectra of the SMScene image.

4) CMScene-sub: This image is a 40 × 70 pixel subset of the CMScene image, as shown in Fig. 5(b). There are four endmembers of moss, pebbles, true vegetation, and false vegetation, and the image does not contain the sticks endmember. The corresponding 640 × 1120 pixel area of the CMScene high-resolution image was used for obtaining the ground-truth abundance maps. The spectra of the four endmembers are the same as the spectra of the CMScene image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 5. SMScene-sub and CMScene-sub mixed images. (a) 45 × 45 mixed image. (b) 40 × 70 mixed image.

TABLE IV
NOE ESTIMATION FOR THE SMSCENE AND CMSCENE IMAGES WITH THE HIGHEST ACCURACIES MARKED IN RED

Data	*Ground truth*	*HySime*	*GENE_AH*			*NWHFC*			*FAHU*
			10^{-4}	10^{-6}	10^{-8}	10^{-3}	10^{-4}	10^{-5}	
SMScene	4	28	46	42	42	5	5	**4**	**4**
CMScene	5	30	41	41	41	4	4	4	**5**

B. Performance Evaluation Indices

For the accuracy assessment, the NOE estimation results were directly compared with the ground-truth NOE, and the EE accuracy was evaluated using the SAD

$$\mathrm{SAD}_i = \arccos \frac{\hat{\mathbf{a}}_i^T \mathbf{a}_i}{\|\hat{\mathbf{a}}_i\|_2 \|\mathbf{a}_i\|_2} \tag{2}$$

where $\mathbf{a}_i$ is the ground truth of the $_{i\text{th}}$ endmember spectra and $\hat{\mathbf{a}}_i$ is the estimated endmember spectra. Note that, in the results of this article, the SAD results are in degrees.

To describe the difference between the estimated abundances and the corresponding ground truth, the root-mean-square error (RMSE) indicator is introduced

$$\mathrm{RMSE}_i = \frac{1}{\sqrt{N}} \left\| \mathbf{S}_i - \hat{\mathbf{S}}_i \right\|_F \tag{3}$$

where $\mathbf{S}_i$ and $\hat{\mathbf{S}}_i$ are the ground truth of the$_{i\text{th}}$ endmember abundance and the estimated abundance of the $_{i\text{th}}$ endmember, respectively.

C. Quantitative Evaluation Experimental Results

In this part, we describe the experiments conducted based on the SMScene and CMScene images for the NOE, EE, and AE tasks.

1) Results for the Estimation of the NOE: By analyzing the results for the SMScene image shown in Table IV, it can be seen that the FAHU algorithm can obtain the correct NOE. It is also apparent that the NOE estimated by GENE-AH decreases with the false alarm probability P_F. For the results for the CMScene image listed in Table IV, overall, the FAHU method obtains the best NOE estimation accuracy in this experiment. The NOE results for GENE-AH and NWHFC are robust with regard to the false alarm probability, but the NOEs are higher than the ground truth. Comparing the FAHU method and the other NOE methods, the main difference is that the FAHU method utilizes the abundance anomaly subspace for determining the NOE, so that FAHU can estimate a relatively accurate NOE.

2) Results for the EE: In this experiment, the MVCNMF and uDAS methods were initialized with VCA. The quantitative results (SAD) for the endmember spectra extraction are listed in Table V.

For the SMScene image, the OSP method obtains the highest accuracy. The FAHU method can estimate more accurate moss, pebbles, and leaves endmembers, as it takes the spatial information into account. The linear sticks endmember is not extracted correctly by the FAHU method, and the MVSA method obtains the best extraction results for this endmember, compared with the other algorithms. CyCU-Net obtains the best extraction results for the moss and pebbles endmembers.

From the EE results for the CMScene image, N-FINDR obtains the best mean SAD values. The FAHU method shows its superiority in extracting the moss and false vegetation endmembers. CyCU-Net obtains the best extraction results for the pebbles endmember. For the linear sticks endmember, the OSP and N-FINDR methods extract the most accurate results. The true vegetation endmember can be extracted correctly by OSP and N-FINDR, and the FAHU method is also capable of extracting a relatively accurate spectrum.

3) Results for the AE: In the AE experiments, MVCNMF and uDAS were initialized in the same way as in the EE experiments. The nonlinear mixed algorithms GBM and PPNMM are initialized with VCA. The abundances for VCA, N-FINDR, and MVSA were estimated by FCLS.

From the SMScene results listed in Table VI, it can be seen that the RMSE values of GBM are the lowest and it estimates the moss abundance more accurately than the other methods. CyCU-Net can estimate the abundance of the pebbles endmember very accurately. The sticks and leaves abundances estimated by N-FINDR and FCLS are the most accurate, and the abundances estimated by VAC and FCLS are also relatively good. The RMSE and abundances of the moss and pebbles estimated by nonlinear unmixing algorithms are more accurate than those of most of the compared algorithms. The proposed FAHU method estimates the abundances of the leaves more accurately than many of the algorithms.

The AE results for the CMScene image are listed in Table VI. The PPNMM method obtains the best overall accuracy in terms of RMSE and also obtains the best results for the moss and pebbles abundances. The nonlinear unmixing algorithms estimate abundances more accurately than other algorithms. The proposed FAHU method can estimate the true vegetation and false vegetation abundances more accurately than the other methods. CyCU-Net estimates the true vegetation abundances better than most of the algorithms.

D. Noise Sensitivity Analysis

The SMScene-sub and CMScene-sub images were used to analyze the sensitivity to noise, with the Gaussian white noise level varied from 10 to 50 dB. The SNR of the data was defined as follows:

$$\mathrm{SNR} = 10\log_{10} \frac{E\left[\mathbf{x}^T \mathbf{x}\right]}{E\left[\varepsilon^T \varepsilon\right]} \tag{4}$$

where $\mathbf{x}$ is the clear hyperspectral image, and ε is the noise.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

TABLE V

EE RESULTS FOR THE SMSCENE AND CMSCENE IMAGES WITH THE TOP-THREE ACCURACIES MARKED IN RED, GREEN, AND BLUE, RESPECTIVELY

Data	Endmember	Method							
		VCA	OSP	N-FINDR	MVSA	MVCNMF	uDAS	CyCU-Net	FAHU
SMScene	Moss	0.2978	0.2027	0.2700	1.5708	0.4307	0.3365	0.1938	0.1643
	Pebbles	0.0419	0.0373	0.0373	0.1195	0.0652	0.0494	0.0391	0.0096
	Sticks	0.0558	0.0623	0.0623	0.0526	0.0992	0.0901	0.3187	0.2909
	Leaves	0.0205	0.0157	0.0157	0.1377	0.0366	0.0449	0.3142	0.0032
	Mean SAD	0.1040	0.0795	0.0963	0.4701	0.1579	0.1302	0.2164	0.1170
CMScene	Moss	0.0674	0.3259	0.1289	0.6873	0.3031	0.1656	0.4659	0.0514
	Pebbles	0.1318	0.1314	0.1314	0.1383	0.3154	0.2929	0.1205	0.5008
	Sticks	0.0400	0.0397	0.0397	1.5708	0.0748	0.2029	0.2266	0.4585
	True vegetation	0.0246	0.0273	0.0273	0.0920	0.1199	0.0875	0.1832	0.0359
	False vegetation	0.2462	0.1035	0.1035	0.7374	0.4610	0.4042	0.1301	0.0634
	Mean SAD	0.1020	0.1256	0.0862	0.6452	0.2745	0.2306	0.2253	0.2220

TABLE VI

AE RESULTS FOR THE SMSCENE AND CMSCENE IMAGES WITH THE TOP-THREE ACCURACIES MARKED IN RED, GREEN, AND BLUE, RESPECTIVELY

Data	Endmember	Method								
		Linear mixing model					Nonlinear mixing model		Deep learning	
		VCA	N-FINDR	MVSA	MVCNMF	FAHU	GBM	PPNMM	uDAS	CyCU-Net
SMScene	Moss	0.9631	0.6246	0.6046	0.4459	0.6586	0.4341	0.4387	0.4626	0.6739
	Pebbles	0.5514	0.5521	0.1025	0.0566	0.0581	0.0403	0.0420	0.0464	0.0326
	Sticks	0.0850	0.0839	0.2144	0.0859	0.4337	0.0885	0.0871	0.0872	0.3894
	Leaves	0.3868	0.2016	0.3350	0.3657	0.3008	0.3622	0.3655	0.3826	0.3903
	RMSE	1.1783	0.8617	0.7309	0.5858	0.8459	0.5736	0.5791	0.6084	0.8713
CMScene	Moss	0.8924	0.8794	0.8161	0.7213	0.5678	0.3408	0.2835	0.4672	0.6154
	Pebbles	0.5021	0.6183	0.1982	0.0908	0.0718	0.0696	0.0681	0.0865	0.0775
	Sticks	0.3337	0.0691	0.1205	0.0816	0.5194	0.0698	0.0770	0.0570	0.5389
	True vegetation	0.2632	0.2506	0.2392	0.5235	0.1498	0.2994	0.2278	0.2591	0.1626
	False vegetation	0.2803	0.4469	0.4834	0.2283	0.1079	0.1498	0.1507	0.2307	0.2350
	RMSE	1.1435	1.1928	1.0054	0.9281	0.7947	0.4878	0.4068	0.5911	0.8699

1) Noise Sensitivity Analysis With the SMScene-sub Image: Gaussian white noise from 10 to 50 dB was added to the SMScene-sub image to compare the performance of the different algorithms. Fig. 6 shows the mean SAD and RMSE values of the algorithms, with the noise level ranging from 10 to 50 dB. The EE results with 30-dB noise are compared with the ground truth of the SMScene-sub image in Fig. 7, and the corresponding abundance maps are shown in Fig. 8.

As the noise level decreases, the mean SAD of all the algorithms generally decreases, and the proposed FAHU method can obtain the best results at all the noise levels. OSP and N-FINDR extract the endmembers well with clear data, but they cannot extract the endmembers well with the noisy images. The mean SAD of MVSA is higher at the 10-dB SNR level than the other noise levels. The mean SAD of uDAS is slightly lower than that of VCA but is very close at all noise levels, but the RMSE of uDAS is much lower than that of VCA and FCLS. The uDAS method performs better than CyCU-Net in EE and AE due to its denoising ability. The proposed FAHU method can estimate relatively accurate abundances, as shown in Fig. 6(b). The EE algorithms using FCLS for AE do not perform as well as the blind unmixing algorithms. From Fig. 7, it can be seen that all the extracted spectral curves are close to the ground-truth endmember spectra, and the spectra obtained by the proposed FAHU method are closer than the others. The spectra extracted by OSP and N-FINDR are jagged, as shown in Fig. 7(c) and (d). From the abundance maps shown in Fig. 8,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

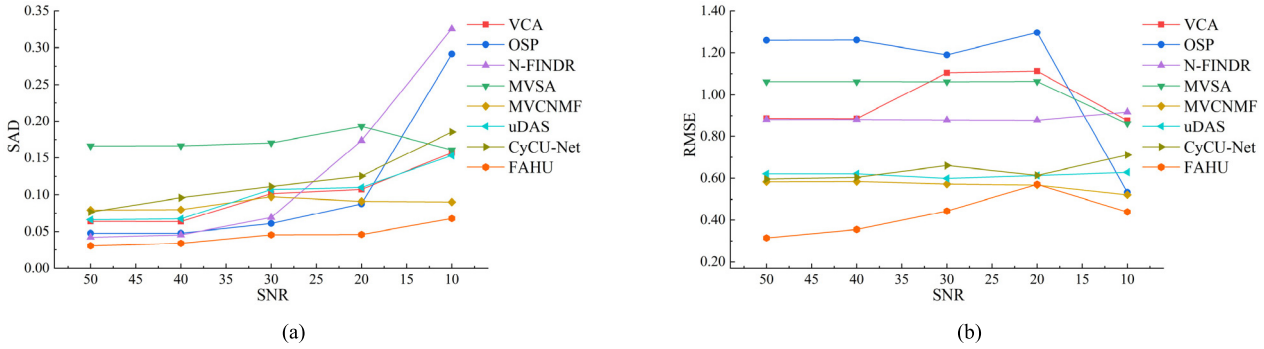


Fig. 6. (a) SAD and (b) RMSE results for every endmember for the SMScene-sub image.

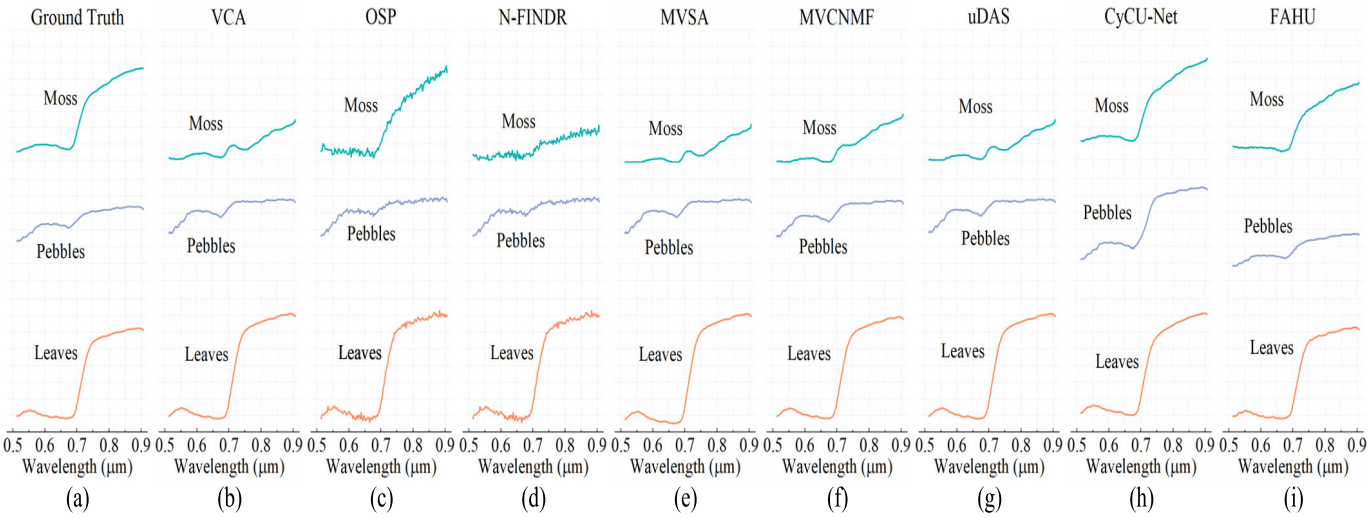


Fig. 7. Comparison of the endmember spectra with the ground truth for the SMScene-sub image with 30-dB noise. (a) Ground truth. (b) VCA. (c) OSP. (d) N-FINDR. (e) MVSA. (f) MVCNMF. (g) uDAS. (h) CyCU-Net. (i) FAHU.

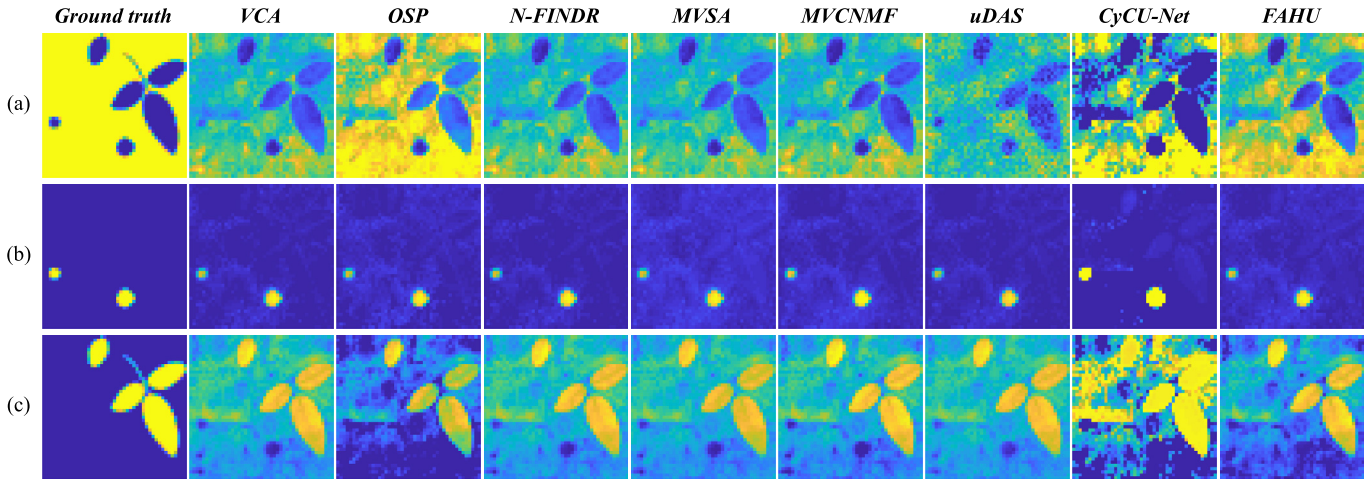


Fig. 8. Comparison of the endmember spectra with the ground truth for the SMScene-sub image. (a) Moss. (b) Pebbles. (c) Leaves.

it can be seen that the main difference between the different algorithms is in the estimation of moss and leaves. However, the proposed FAHU method can estimate these abundances well.

2) Noise Sensitivity Analysis for the CMScene-Sub Image: Gaussian white noise from 10 to 50 dB was added to the CMScene-sub image at 10-dB intervals. The mean SAD and RMSE results are shown in Fig. 9. The extracted endmembers

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

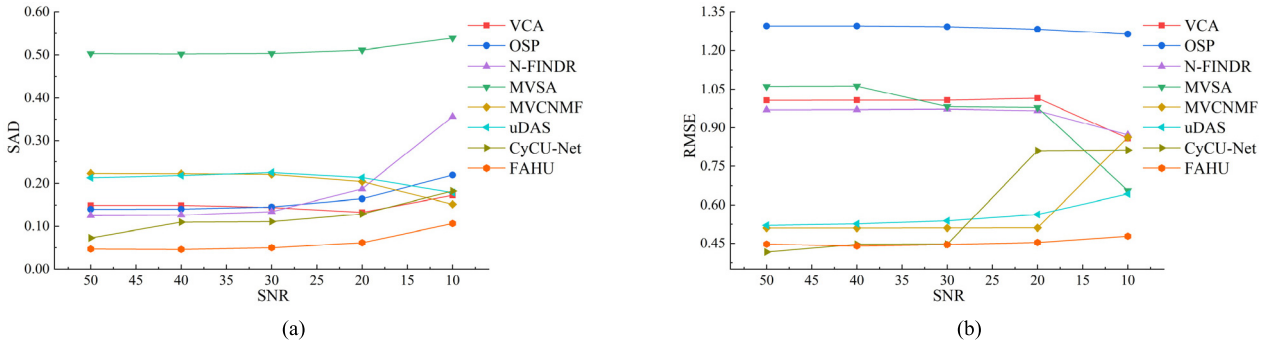


Fig. 9. (a) SAD and (b) RMSE results for every endmember for the CMScene-sub image.

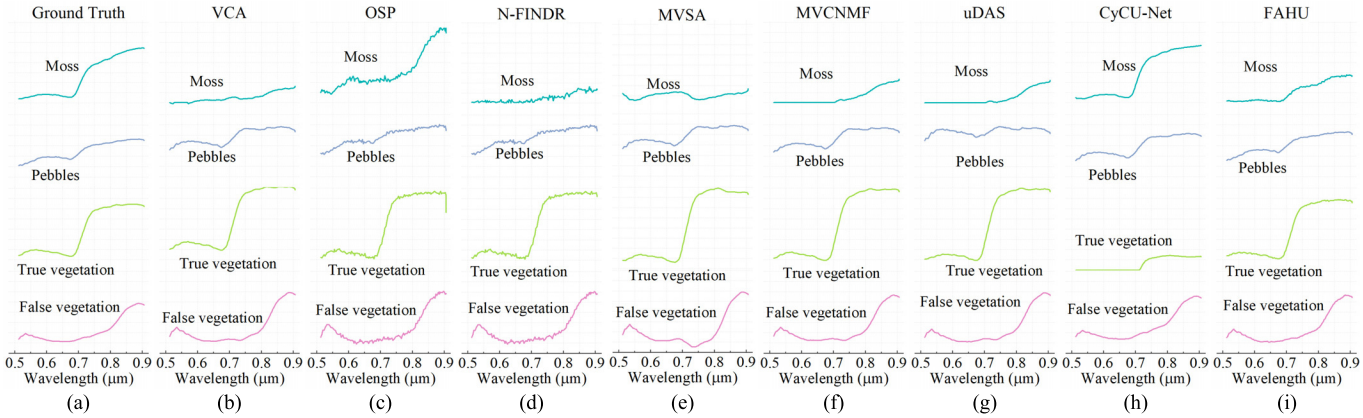


Fig. 10. Comparison of the endmember spectra with the ground truth for the CMScene-sub image with 30-dB noise. (a) Ground truth. (b) VCA. (c) OSP. (d) N-FINDR. (e) MVSA. (f) MVCNMF. (g) uDAS. (h) CyCU-Net. (i) FAHU.

obtained with 30-dB SNR are compared with the ground truth of the CMScene-sub image in Fig. 10. The corresponding AE results are shown in Fig. 11.

As the noise level decreases, the accuracy of the EE algorithms increases and the AE accuracy also increases, as shown in Fig. 9(a) and (b), respectively. The proposed FAHU method obtains better EE results than the other algorithms. CyCU-Net also performs well in EE. The reason for the mean SAD of MVCNMF and uDAS increasing is that the moss EE accuracy decreases a lot. For the AE, the blind unmixing algorithms perform better than the two-step unmixing algorithms. The compared spectral curves show that the endmember spectra extracted by OSP, N-FINDR, and CyCU-are not smooth enough. The true vegetation endmembers extracted by CyCU-Net and FAHU have some small values, and the AE for the true vegetation is not good enough, as shown in Fig. 11. The MVSA method extracts a spectra for true vegetation that is very different from the ground-truth spectra, and the corresponding abundance map is also poor. The moss and true vegetation abundances are mistakenly estimated by all the methods. The false vegetation is mistakenly estimated as pebbles. The uDAS method estimates a noisy abundance map for the moss endmember.

E. Discussion

In this part, a comparison based on the SMScene and CMScene images between the two FAHU algorithms—SPICE and FAHU—is presented. Table VII lists the results for the two algorithms. As SPICE estimates a larger NOE than the ground truth, zero values for the endmember spectra and abundances maps are added for the quantitative analysis.

TABLE VII
COMPARISON BETWEEN SPICE AND FAHU FOR THE NOE,

Data	Scientific problem	Ground truth	SPICE	FAHU
SMScene	***NOE***	4	5	4
	EE	-	0.4125	0.1170
	AE	-	0.9783	0.8459
CMScene	***NOE***	5	7	5
	EE	-	0.5365	0.2220
	AE	-	0.9305	0.7947

For the SMScene image, we can find that the proposed FAHU method obtains a more accurate NOE, and SPICE estimates a larger number than the ground truth. SPICE adds a zero-mean Laplacian distribution to promote the sparsity, and as the simulated scene is both homogeneous and flat, the spectral variability increases the difficulty of SPICE obtaining a relatively accurate NOE, so that the estimated NOE is higher than the ground truth. Based on this, the EE and AE results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

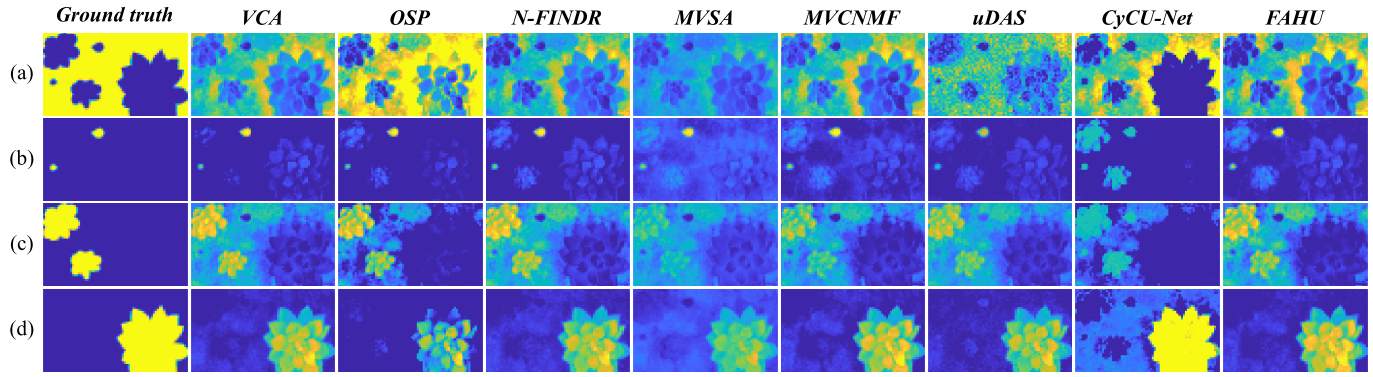


Fig. 11. Comparison of the abundance maps for the CMScene-sub image with 30-dB noise. (a) Moss. (b) Pebbles. (c) True vegetation. (d) False vegetation.

of SPICE are inferior to those of FAHU, which shows that an accurate NOE is of vital importance for FAHU. In the CMScene image, the results allow a similar conclusion.

For the SMScene image, for the endmember spectra extraction, the proposed FAHU method does not obtain the best results, especially for the linear feature, i.e., sticks, which is because it is based on the pure pixel assumption. As a result, the weak targets (with less than one pixel) are ignored as anomalies, so that the endmember spectrum of sticks is not accurate enough.

In the CMScene, for the linear feature, i.e., sticks, the proposed FAHU method cannot estimate this well, allowing a similar conclusion to that for the SMScene experiment. However, for the endmember spectra estimation of moss and false vegetation, FAHU does obtain the best accuracy as it has the advantage in estimating the endmembers with larger proportions in the scene. For the abundance map inversion, FAHU can obtain better results than SPICE.

The proposed FAHU algorithm has the advantage in estimating an accurate NOE, which is the most important problem of hyperspectral unmixing and in turn affects the accuracy of the EE and AE. For the materials with larger proportions, FAHU can estimate more accurate endmember spectra, but for the materials with less spatial information to learn, it cannot obtain good results.

V. Conclusion

The mixed pixel phenomenon occurs everywhere in Earth surface systems. Remote sensing is an ideal data source for large-scale monitoring, but mixed pixels that contain more than one material type are common in mid/low spatial resolution remote sensing imagery. The mixing mechanisms can be categorized into two main categories: linear and nonlinear mixing models. In order to deal with the mixed pixel problem in hyperspectral imagery, a number of state-of-the-art methods have been proposed to solve this problem, which contains three main parts: NOE estimation, EE, and AE. Nevertheless, for the datasets used to test the algorithms, there are some common problems, in that it is difficult to simulate the mixing in real scenes, it is often not easy to obtain the ground-truth reference, and the scenes are generally simple when compared with real distributions.

In this study, in order to simulate the real mixing process, the RMMS dataset was built, which can be used to quantitatively evaluate the unmixing accuracy of different algorithms. The RMMS dataset consists of a simple mixture scene with homogeneous flat materials and a complex mixture scene with the absence of pure sticks pixels. The features of the RMMS dataset also take the point, line, and polygon characteristics into consideration, and the spectral similarity of the materials increases the challenge of the spectral unmixing. The RMMS dataset provides standard data for analyzing the performance of hyperspectral unmixing. The RMMS dataset can be used to conduct benchmark experiments using the typical unsupervised hyperspectral unmixing algorithms. The proposed FAHU algorithm can also achieve FAHU, while obtaining accurate results. In our future work, the spectral variability will be considered for spectral unmixing, and a new dataset for linear or nonlinear mixing verification will also be built.

References

[1] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[2] N. Wang, W. Li, R. Tao, and Q. Du, "Graph-based block-level urban change detection using Sentinel-2 time series," *Remote Sens. Environ.*, vol. 274, Jun. 2022, Art. no. 112993.

[3] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Jan. 2002.

[4] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sensing*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[5] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[6] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

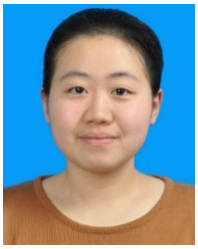
[7] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," in *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.

[8] J. Li, A. Agathos, D. Zaharie, J. M. Bioucas-Dias, and X. Li, "Minimum volume simplex analysis: A fast algorithm for linear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5067–5082, Apr. 2015.

[9] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.

[10] C.-I. Chang and D. Heinz, "Constrained subpixel target detection for remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1144–1159, May 2000.
[11] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.
[12] X. Wang, Y. Zhong, C. Cui, L. Zhang, and Y. Xu, "Autonomous endmember detection via an abundance anomaly guided saliency prior for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2336–2351, Mar. 2021.
[13] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.
[14] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.
[15] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.
[16] A. Zare and P. Gader, "Sparsity promoting iterated constrained endmember detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 3, pp. 446–450, Jul. 2007.
[17] M. Berman, H. Kiiveri, R. Lagerstrom, A. Ernst, R. Dunne, and J. F. Huntington, "ICE: A statistical approach to identifying endmembers in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 10, pp. 2085–2095, Oct. 2004.
[18] M. D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.
[19] E. M. T. Hendrix, I. Garcia, J. Plaza, G. Martin, and A. Plaza, "A new minimum-volume enclosing algorithm for endmember identification and abundance estimation in hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 7, pp. 2744–2757, Jul. 2012.
[20] F. Zhu, "Hyperspectral unmixing: Ground truth labeling, datasets, benchmark performances and survey," 2017, *arXiv:1708.05125*.
[21] G. Swayze, R. N. Clark, K. Kruse, S. Sutley, and A. Gallagher, "Ground-truthing AVIRIS mineral mapping at Cuprite, Nevada," in *Proc. Summaries 3rd Annu. JPL Airborne Geosci. Workshop*, vol. 1, 1992, pp. 47–49.
[22] S. Cook, J. Harsanyi, and V. Faber, "Evaluation of algorithms for compressing hyperspectral data," *Proc. SPIE*, vol. 5234, pp. 712–717, Feb. 2004.
[23] A. Mueller, A. Hausold, and P. Strobl, "HySens-DAIS/ROSIS imaging spectrometers at DLR," *Proc. SPIE*, vol. 4545, pp. 225–235, Jan. 2002.
[24] M. Zhao, J. Chen, and Z. He, "A laboratory-created dataset with ground truth for hyperspectral unmixing evaluation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2170–2183, Jul. 2019.
[25] A. M. Ambikapathi, T. H. Chan, C. Y. Chi, and K. Keizer, "Hyperspectral data geometry-based estimation of number of endmembers using p-norm-based pure pixel identification algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2753–2769, May 2013.
[26] D. Shah, T. Zaveri, Y. N. Trivedi, and A. Plaza, "Entropy-based convex set optimization for spatial–spectral endmember extraction from hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4200–4213, Jul. 2020.
[27] T.-H. Chan, C.-Y. Chi, Y.-M. Huang, and W.-K. Ma, "A convex analysis-based minimum-volume enclosing simplex algorithm for hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 57, no. 11, pp. 4418–4432, Nov. 2009.
[28] M. D. Craig, "Minimum-volume transforms for remotely sensed data," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 3, pp. 542–552, May 1994.
[29] N. Raksuntorn and Q. Du, "Nonlinear spectral mixture analysis for hyperspectral imagery in an unknown environment," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 836–840, Oct. 2010.
[30] A. Halimi, Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Nonlinear unmixing of hyperspectral images using a generalized bilinear model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4153–4162, Nov. 2011.
[31] Y. Altmann, A. Halimi, N. Dobigeon, and J.-Y. Tourneret, "Supervised nonlinear spectral unmixing using a postnonlinear mixing model for hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 21, no. 6, pp. 3017–3025, Jun. 2012.
[32] C.-I. Chang, W. Xiong, H.-M. Chen, and J.-W. Chai, "Maximum orthogonal subspace projection approach to estimating the number of spectral signal sources in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 504–520, Jun. 2011.
[33] O. Kuybeda, D. Malah, and M. Barzohar, "Rank estimation and redundancy reduction of high-dimensional noisy signals with preservation of rare vectors," *IEEE Trans. Signal Process.*, vol. 55, no. 12, pp. 5579–5592, Dec. 2007.
[34] H. Ren and C.-I. Chang, "Automatic spectral target recognition in hyperspectral imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 4, pp. 1232–1249, Oct. 2003.
[35] X. Wang, Y. Zhong, Y. Xu, L. Zhang, and Y. Xu, "Saliency-based endmember detection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3667–3680, Jul. 2018.
[36] X. Li, R. Huang, and L. Zhao, "Correntropy-based spatial–spectral robust sparsity-regularized hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1453–1471, Feb. 2021.
[37] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Mar. 2022, Art. no. 5503914.
[38] A. Min, Z. Guo, H. Li, and J. Peng, "JMNet: Joint metric neural network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Apr. 2022, Art. no. 5505412.



Chunyang Cui received the B.S. degree in surveying and mapping engineering from Northeastern University, Shenyang, China, in 2018. She is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China.

Her major research interests include the hyperspectral unmixing and blind source separation.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has authored more than 100 research articles in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He received the Second-Place Prize in the 2013 IEEE GRSS Data Fusion Contest and the Single-view Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.



Xinyu Wang received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, in 2019.

Since 2019, he has been an Associate Research Fellow with the School of Remote Sensing and Information Engineering, Wuhan University. His major research interests include hyperspectral unmixing, detection, restoration, and tracking.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He is currently a "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China at the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He was a Principal Scientist for the China State Key Basic Research Project (2011–2016) appointed by the Ministry of National Science and Technology of China to lead the remote sensing program in China. He has authored more than 700 research articles and five books. He is the Institute for Scientific Information (ISI) highly cited author. He is the holder of 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams received the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He is the Founding Chair for the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter. He also serves as an associate editor or an editor for more than ten international journals. He is currently serving as an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:56:15 UTC from IEEE Xplore. Restrictions apply.